Prognostic Confounds and the Coupling Ceiling: A Silent Repair Artifact and Its Analytic Boundary

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Purpose. Two paper 02 simulation jobs (jobs 28 and 30) reported that every analysis specification, `G`-family and `F`-family alike, lost 15 to 25 points of power when a strong prognostic confound was added to the data-generating process alongside the true biomarker-treatment interaction. This note shows that the loss was not a property of any analysis model but of the data-generating process itself: the confound strength used was infeasible, the joint covariance matrix was not positive definite, and the drivers' silent repair step removed part of the interaction signal before any model saw the data. It then derives, in closed form, the exact boundary of the feasible region for a prognostic coupling alongside the predictive coupling c_{bm}, so that future stress cells can be chosen inside it. Both halves are verified by code in the repository. The studies are exploratory.

1. Summary

- At confound strength 0.6 on the prog-PB and prog-TV-const pathways (and prog-BL at $c_{bm} = 0.45$), the joint correlation matrix of the refined-process DGP is non-positive-definite on every Hybrid path, with smallest eigenvalue between -0.11 and -0.85.
- The paper 02 drivers ran with silent repair enabled, so `corpcor::make.positive.definite` replaced the matrix by a nearby positive-definite one. The repair shrinks the population on-drug/off-drug slope contrast of the outcome on the biomarker, which is the quantity the interaction term estimates, from -3.5 to -2.4 (prog-PB) and from -3.3 to -1.8 (prog-TV-const). The fitted estimates from the simulations match these repaired values, and standard errors do not grow.
- No analysis model can recover a signal the DGP did not generate. Job 30's three random-effects augmentations of `G9` changed power by at most one point in every cell, which is the expected result once the mechanism is understood.
- The compendium's Schur-complement ceiling for c_{bm} extends directly to a biomarker row with several couplings. The feasible region is the interior of an ellipsoid $c^\top A c < 1$, with A computed from the inverse of the correlation matrix with the biomarker removed. At $c_{bm} = 0.45$ on Hybrid, the largest feasible prog-PB or prog-TV-const coupling is 0.306 and the largest prog-BL coupling is 0.496; at $c_{bm} = 0.30$ they are 0.451 and 0.815. A coupling of 0.6 on PB or TV is infeasible at every c_{bm}.

- Consequences: job 28's strength-0.6 power numbers must not be cited as a confound effect; every valid cell (strength 0.3) shows no power cost at all; the project's non-PD default is now stop, not repair, in every collection; and stress grids should be chosen from the ceiling table in Section 5 and checked before submission.

2. Background

2.1 The prognostic pathways

Paper 01's Study 3 and `docs/57` Section 7.3 define four ways a biomarker can be prognostic without being predictive, each a separate channel in the DGP:

Pathway	Parameter	Mechanism
prog-BL	`c.bm.bl` (c_{bl})	correlation between the biomarker and baseline severity BL
prog-TV-const	`c.bm.tv` (c_{tv})	correlation between the biomarker and the natural-history component TV at every timepoint
prog-TV-grow	`b.bm.tv` (b_{tv})	an additive shift of the TV mean proportional to the standardized biomarker
prog-PB	`c.bm.pb` (c_{pb})	correlation between the biomarker and the placebo-response component PB at every timepoint

Three of the four are entries in the biomarker row of the joint correlation matrix. prog-TV-grow is a mean shift and never touches the covariance; it is absorbed exactly by the biomarker main effect and plays no part in what follows.

2.2 The predictive coupling and its ceiling

Under covariance moderation (the refined process), the biomarker-treatment interaction is encoded as a correlation c_{bm} between the biomarker and the drug-response component BR, equal to c_{bm} on drug and decaying off drug. `docs/34` and the mathematics compendium show that c_{bm} is bounded above by a ceiling c^*_{bm} beyond which the joint matrix is not positive definite, and give the ceiling

in closed form by a Schur complement on the biomarker row. Every production grid in papers 01 and 02 sits below it.

2.3 Silent repair

Hendrickson's published code repairs a non-positive-definite covariance matrix with `corpcor::make.positive.definite(tol = 1e-3)` and continues. The refined process (paper 01, Table `tab:spec` and Appendix B.2) recommends stopping instead. A stop option (`stopOnNonPD` in the package, `stop_on_nonpd` in the tidyverse collection) was added on 2026-10-21 with a backward-compatible default of `FALSE`, so every caller that did not opt in kept repairing silently. The paper 02 drivers did not opt in.

3. The issue: what jobs 28 and 30 actually measured

3.1 Job 28, the prognostic stress test

Driver `28-run-g9-prognostic-stress.R` crossed the four pathways with confound strength $\{0.3, 0.6\}$, $c_{bm} \in \{0, 0.45\}$ and $N \in \{35, 70\}$ on the Hybrid design, 500 replicates per cell, fitting `G1-G9`, `E0` and `F1`/`F1A`/`F1A0`/`F1A0B` to the same simulated trial each replicate. Its two questions were whether a doubled confound inflates `G9`'s Type I error and whether a confound co-occurring with a true interaction costs power.

Type I error was nominal everywhere: `G9` rejected between 0.042 and 0.050 at all sixteen null cells. Power at $c_{bm} = 0.45$ (`G9`, $N = 35$ / $N = 70$):

Pathway	No confound	Strength 0.3	Strength 0.6
prog-BL	0.759 / 0.969	0.782 / 0.976	0.738 / 0.950
prog-PB	0.759 / 0.969	0.798 / 0.968	0.597 / 0.853
prog-TV-const	0.759 / 0.969	0.782 / 0.968	0.502 / 0.772
prog-TV-grow	0.759 / 0.969	0.778 / 0.970	0.778 / 0.970

(The no-confound baseline is job 30's, 1000 replicates.) At strength 0.3 every pathway is at or above baseline within Monte Carlo error. At 0.6 the PB and TV-const pathways drop by 17 to 26 points, and so does every other specification in the job: F1A0B, for instance, falls from 0.741 to 0.562 on prog-PB at $N = 35$. The uniformity across fourteen specifications of three different families was the first hint that the loss was upstream of the analysis.

3.2 Job 30, a hypothesis that was wrong

The initial reading of job 28 was that the confound added patient-level variance that the models' random-intercept structure could not absorb. Driver `29-run-g9-random-effects.R` therefore fit `G9` with three richer random-effects structures (random slope on time, on expectancy, and both) on the same twelve cells at 1000 replicates. None recovered anything: all three variants matched plain `G9` to within one point of power and of standard error in every cell, with paired per-replicate differences of at most 0.008, at the cost of up to 2.5% non-convergence.

The per-cell estimates in the same output already contradicted the variance hypothesis. The mean `G9` estimate of the interaction at $N = 35$ was -3.61 with no confound, -2.50 under prog-PB 0.6 and -1.97 under prog-TV-const 0.6, while the empirical standard error fell from 1.38 to 1.16 to 1.02. An added-variance mechanism would leave the estimate centered and inflate its spread; what was observed was attenuation with a tighter spread.

3.3 The direct check

Script `30-check-prognostic-confound-pd.R` builds the covariance matrix for every (pathway, strength, c_{bm}) combination the jobs used, on each of the four Hybrid paths, with no simulation, and reports two things: whether the raw matrix is positive definite (`build_sigma_matrix()`'s own `was_pd_corrected` flag and the smallest eigenvalue of DRD with D the standard deviations), and the population slope of the outcome on the biomarker at each timepoint, computed from the matrix directly. Since the outcome is $S_t = \mathrm{BL} - TV_t - PB_t - BR_t$,

$$\beta_t = \frac{\mathrm{Cov}(bm, \mathrm{BL}) - \mathrm{Cov}(bm, TV_t) - \mathrm{Cov}(bm, PB_t) - \mathrm{Cov}(bm, BR_t)}{\mathrm{Var}(bm)},$$

and the on-drug minus off-drug average of β_t is the population value of the interaction contrast. Comparing it before and after repair isolates what the repair did to the signal.

Pathway	Strength	c_{bm}	Paths repaired	Min. eigen-value	Contrast, raw	Contrast, repaired
none	0	0.45	0	0.180	−3.26	−3.26
prog-BL	0.3	0.45	0	0.145	−3.26	−3.26
prog-BL	0.6	0.45	4	−0.110	−3.26	−3.00
prog-PB	0.3	0.45	0	0.010	−3.40	−3.40
prog-PB	0.6	0	4	−0.240	−0.27	−0.24
prog-PB	0.6	0.45	4	−0.642	−3.53	−2.38

Pathway	Strength	c_{bm}	Paths repaired	Min. eigen-value	Contrast, raw	Contrast, repaired
prog-TV-const	0.3	0.45	0	0.011	−3.26	−3.26
prog-TV-const	0.6	0	4	−0.354	0.00	0.00
prog-TV-const	0.6	0.45	4	−0.853	−3.26	−1.81
prog-TV-grow	0.6	0.45	0	0.180	−3.26	−3.26

Three things follow. First, every strength-0.3 cell is positive definite, which is why no power was lost there. Second, the two strength-0.3 cells on PB and TV-const have a smallest eigenvalue of 0.010, almost exactly on the boundary; Section 5 shows why. Third, the repaired contrasts, −2.38 and −1.81, are what the fitted models reported (−2.50 / −2.32 and −1.97 / −1.84 at $N = 35$ / 70). The power loss is the repair removing 33% to 45% of the interaction from the generated data. There was nothing for any analysis model to recover.

3.4 Two side observations

prog-PB is not a pure confound of the interaction. Even at $c_{bm} = 0$, the raw contrast under prog-PB is nonzero (−0.14 at strength 0.3, −0.27 at 0.6, against −3.3 at the alternative). The placebo-response amplitude scales with expectancy, which in the Hybrid design differs between on-drug and off-drug visits, so a biomarker correlated with PB acquires a small on/off slope difference by construction. `G9` showed no detectable Type I error inflation from it at $N \leq 70$, but this is exactly the mechanism the `F`-family's expectancy term exists to absorb, and it may matter at larger N or stronger couplings.

Zero-variance components are a separate case. Flipping the repair default (Section 6) exposed a second, unrelated use of the repair. Because PB's amplitude scales with expectancy, a timepoint with expectancy 0 (a run-in phase) has exactly zero PB variance and makes the covariance matrix singular by construction. That is a degenerate component, not an infeasible parameter set. The stop path now checks and factors only the positive-variance sub-block and draws the degenerate components at their means; no correlation is modified.

4. The analytic boundary

4.1 Setup

For one participant on one path with n visits, the DGP draws $(B, \mathrm{BL}, TV_{1:n}, PB_{1:n}, BR_{1:n})$ from a multivariate normal distribution. Write R for its correlation matrix and R_0 for R with the biomarker row and column removed. R_0 contains the within-component AR(1) blocks, the cross-component correlations and the uncorrelated BL, and depends on no biomarker coupling. The biomarker row is a sum of fixed patterns scaled by the couplings:

$$r = c_{bm}\, g + c_{pb}\, p + c_{tv}\, q + c_{bl}\, b,$$

where g is the predictive pattern (1 on the BR positions while on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after first exposure, 0 before exposure, 0 elsewhere), p is 1 on every PB position, q is 1 on every TV position and b is 1 on BL. The variances rescale R to the covariance and do not affect positive definiteness.

4.2 The condition

By the Schur complement on the biomarker row, $R \succ 0$ if and only if $R_0 \succ 0$ and

$$r^\top R_0^{-1} r < 1.$$

The left side is the biomarker's squared multiple correlation on all other variables, so the condition says only that the biomarker cannot be more than perfectly predictable from the rest. Substituting the pattern expansion,

$$r^\top R_0^{-1} r = c^\top A\, c, \qquad A_{jk} = x_j^\top R_0^{-1} x_k, \quad x \in \{g, p, q, b\},$$

with $c = (c_{bm}, c_{pb}, c_{tv}, c_{bl})$. The feasible region is the interior of the ellipsoid $c^\top A c < 1$ in the positive orthant. The compendium's single-coupling ceiling is the special case $c_{bm}^* = A_{gg}^{-1/2}$.

4.3 The boundary for one coupling given the others

Fix all couplings but one, say c_{pb}. Then $c^\top A c = 1$ is a quadratic in c_{pb},

$$A_{pp}\, c_{pb}^2 + 2\Big(\sum_{k \neq p} A_{pk} c_k\Big) c_{pb} + \Big(\sum_{j,k \neq p} c_j A_{jk} c_k - 1\Big) = 0,$$

and the ceiling is its positive root. If the discriminant is negative the fixed couplings are already infeasible on their own. A design's ceiling is the minimum over its paths, since all paths share one parameter set.

Because BL is uncorrelated with every response component, b is an eigenvector of R_0 with $A_{bb} = 1$ and $A_{bk} = 0$ for $k \neq b$, so c_{bl} enters only as c_{bl}^2: the prog-BL ceiling given c_{bm} is $\sqrt{1 - c_{bm}^2 A_{gg}}$. The PB and TV patterns are structurally identical (a constant on every position of one component with the same AR(1) block and the same cross-component correlations), so $A_{pp} = A_{qq}$ and $A_{gp} = A_{gq}$, and prog-PB and prog-TV-const have identical ceilings. Expectancy scaling of PB lives in the standard deviations and does not enter.

4.4 Implementation and verification

Script `31-prognostic-coupling-ceiling.R` obtains R_0 and the four patterns from the paper 02 driver's own `build_sigma_matrix()` (building the correlation matrix at zero couplings, then at each unit coupling in turn), forms A per path, and solves the quadratic. Each closed-form ceiling is checked against the smallest eigenvalue of the full R: positive at 0.999 times the ceiling, negative at 1.001 times it. All fourteen cases in the table below pass.

5. Ceilings for the paper 02 configuration

Hybrid design, minimum over its four paths, $t_{1/2} = 0.5$ week, $\rho = 0.7$, $c_1 = 0.2$ at every lag (the corrected constants of `docs/61`).

Coupling	Alone	Given $c_{bm} = 0.30$	Given $c_{bm} = 0.45$
c_{bm}	0.518		
c_{pb}	0.513	0.451	0.306
c_{tv}	0.513	0.451	0.306
c_{bl}	1.000	0.815	0.496

In the other direction, given c_{pb} or $c_{tv} = 0.3$ the ceiling on c_{bm} is 0.454; given 0.6, no c_{bm} is feasible.

The coefficient matrix for path 1 (the others differ in the third decimal) is

$$A = \begin{pmatrix} 3.725 & -0.400 & -0.400 & 0 \\ -0.400 & 3.797 & -0.633 & 0 \\ -0.400 & -0.633 & 3.797 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

in the order (g, p, q, b). The negative off-diagonals mean the couplings are mildly complementary rather than purely competing: the response components are

positively correlated with one another, so a biomarker tied to BR already implies a little correlation with PB and TV, and an explicit c_{pb} of the same sign uses up less of the budget than it would if the components were independent.

Reading job 28 against this table: at $c_{bm} = 0.45$, strength 0.3 on PB or TV-const sat at 0.306, inside the region by 0.006, which is the eigenvalue of 0.010 in Section 3.3; strength 0.6 was outside it at every c_{bm}; and prog-BL at 0.6 was outside it at $c_{bm} = 0.45$ (ceiling 0.496) but inside at $c_{bm} = 0$. The c_{bm}-alone ceiling of 0.518 is higher than the 0.480 Hybrid figure in `docs/34` because that figure used the pre-`docs/61` cross-lag constant of 0.1 and $t_{1/2} = 0$.

6. Consequences and recommendations

1. **Do not cite job 28's strength-0.6 power numbers as a confound effect.** The valid finding from jobs 28 and 30 is narrower and cleaner: `G9` holds nominal size under every prognostic pathway at both strengths, and at every feasible strength tested it loses no power relative to the no-confound alternative. The job 30 random-effects variants had no artifact-free motivation and are not a direction worth pursuing.
2. **Non-positive-definite matrices are now refused, never repaired, by default** (commit `4aa4d0ef4`): in the package, the tidyverse and original-extended collections, the nof1power engine, and paper 02's driver core, where a refused cell appears as a non-converged replicate. Silent repair survives only as an explicit `FALSE` opt-in for the scripts that study the repair itself and for reproducing runs made before 2026-10-26. `implementations/original` is untouched, since silent repair is part of the published process it reproduces.
3. **The stop is a backstop, not a design tool.** Parameters must be chosen inside the feasible region before a run, from the table in Section 5 or by running script 31 for a new configuration. A pre-flight eigenvalue check over the whole grid, as the package's `generateSimulatedResults()` already performs, should be added to the paper 02 drivers so an infeasible cell is rejected before the simulation host spends time on it.
4. **A valid "stronger confound" test is still open.** At $c_{bm} = 0.45$ there is almost no room above 0.3. The question can be asked at $c_{bm} = 0.30$, where confounds up to 0.45 are feasible on PB and TV-const and up to 0.81 on BL, or at $c_{bm} = 0.45$ with prog-BL up to 0.49. Whether a ladder of this kind is worth running depends on whether the manuscript needs to claim robustness beyond strength 0.3; it currently does not.
5. **The prog-PB expectancy mechanism (Section 3.4) deserves a targeted size check** at larger N before `G9`'s robustness to a prognostic placebo responder is stated without qualification.

7. Reproducibility

- `analysis/scripts/carryover-sensitivity/30-check-prognostic-confound-pd.R` writes `pd-check.rds` in `analysis/data/carryover-sensitivity/g9-prognostic-stress/` (Section 3.3).
- `analysis/scripts/carryover-sensitivity/31-prognostic-coupling-ceiling.R` writes `.../g9-prognostic-stress/coupling-ceilings.rds` (Section 5).
- Job 28: `28-run-g9-prognostic-stress.R`, results in `.../g9-prognostic-stress/` (commit `a69050073`). Job 30: `29-run-g9-random-effects.R`, results in `.../g9-random-effects/` (commit `3d1ee857a`). Both were generated under the old repair default; their strength-0.6 rows are the artifact described here and are retained for the record.
- The stop-by-default change and the zero-variance handling: commit `4aa4d0ef4`; tests in `inst/tinytest/test_sigma.R`.
- Both scripts run from the repository root in seconds on the host R session and need no simulation.

8. Caveats, unsolicited

- The ceilings in Section 5 are for one configuration (Hybrid, $t_{1/2} = 0.5$, $\rho = 0.7$, $c_1 = 0.2$). They move with every one of those settings; the compendium shows the c_{bm} ceiling alone varies from 0.25 to 0.52 across the project's constructs. Script 31 must be rerun for any other configuration.
- The attenuation argument in Section 3.3 is a population calculation. It explains the direction and size of the observed loss and matches the fitted means, but the simulated estimates are also subject to the usual finite-sample behavior of `G9`'s AIC-selected half-life and CR2 standard errors, which this note does not separate out.
- `G9`'s nominal Type I error at the strength-0.6 null cells was measured under the repaired matrix, so it is a result about a modified DGP, not about the intended one. The strength-0.3 null results are unaffected.
- The claim that no analysis model can recover the lost signal is exact for the population contrast but was checked empirically only for `G9` and its three random-effects variants, and for the specifications in job 28; it is not a theorem about every conceivable estimator of a different estimand.
- Parity tests between the implementation collections were not rerun after the default change.

References

- `analysis/report/compendium-mathematics.tex`, section "Validity of covariance binding: the coupling ceiling", equation `ce-ceiling`.

- `docs/34-cbm-feasible-range.md` and `docs/35-separable-response-covariance.md`, the single-coupling ceiling and its dependence on the correlation construct.
- `docs/57-data-chosen-carryover-half-life.md`, Section 7.3, the four prognostic pathways.
- `docs/60-trials-submission-gap-analysis.md`, 2026-10-26 update, the chronology of jobs 28 and 30 and the default change.
- `docs/61-cross-factor-correlation-bug-paper02-driver.md`, the $c_1 = 0.2$ correction that the ceilings in Section 5 assume.
- `docs/64-f1a0b-first-pass-evaluation.md`, the strength-0.3 prognostic cells for the `F`-family, which Section 3.3 confirms are feasible.
- `analysis/scripts/quick-sim/cbm-ceiling/01-cbm-ceiling.R`, the single-coupling ceiling script this note's script 31 generalizes.